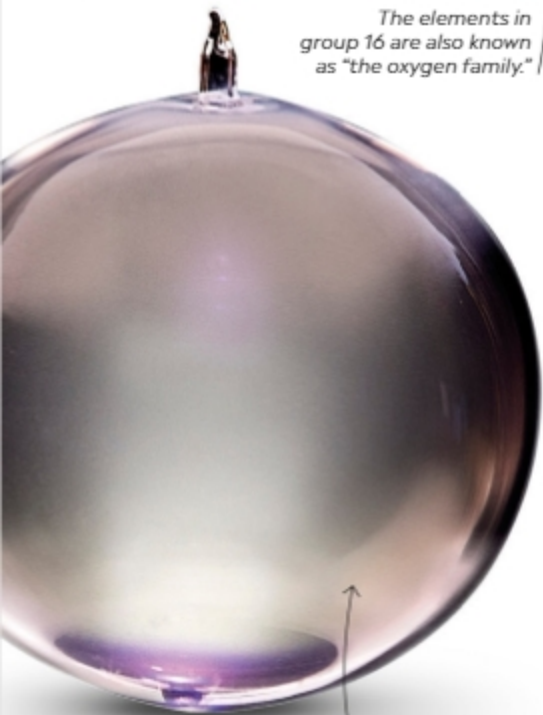


The periodic table shows elements grouped by color. Group 16 elements are highlighted in orange: Oxygen (O), Sulfur (S), Selenium (Se), Tellurium (Te), Polonium (Po), and Livermorium (Lv). A bracket on the left side of the table points to this group.

The elements in group 16 are also known as "the oxygen family."



This glass sphere stores pure oxygen, which produces a silver-blue glow when electrified.



The Oxygen Group

These elements are sometimes called "chalcogens," a combination of the Greek words *khalkós*, meaning "copper" and *genēs*, meaning "born" because they are often found in copper ores. Oxygen, sulfur, and selenium are non-metals, tellurium is a metalloid, and polonium is a metal. Only a few atoms of livermorium have ever been made.

Oxygen



The third-most-abundant element in the universe, oxygen makes up 21 percent of the air we breathe. This colorless gas becomes a blue liquid at -297°F (-183°C), which is used in the medical industry to preserve tissue samples. Oxygen has several forms, including ozone, which forms a protective barrier between Earth and the damaging ultraviolet rays of the sun.

Oxygen reaction

A chemical reaction called oxidation occurs when oxygen comes into contact with any substance. For example, a cut apple turns brown because chemicals in its cells react with oxygen in the air. Rusting is a type of oxidation where iron reacts with oxygen and water to form flaky hydrated iron oxides that weaken the metal.



This apple's brown flesh is a sign of oxidation.

Oxygen cycle

Oxygen is essential for life, because all living things respire—use oxygen to release chemical energy from fuels such as sugars. However, while animals breathe in oxygen and breathe out carbon dioxide, plants take in carbon dioxide and, using sunlight and water, produce oxygen in a process called photosynthesis. They use some of this oxygen, but most is released back into the atmosphere, creating a cycle.